

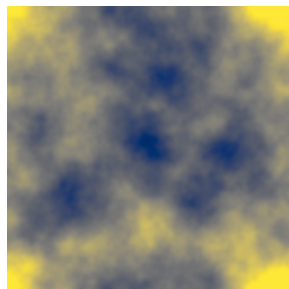
		abiotic selection toward a local optimum	biotic selection the host evades, the parasite follows	dispersal offspring settle near their parents	random genetic drift space-time white noise
host	$\frac{\partial \bar{z}_H}{\partial t} =$	$-A_H \bar{z}_H$	$-B_H (\bar{z}_P - \bar{z}_H)$	$+ \frac{\sigma_H^2}{2} \nabla^2 \bar{z}_H$	$+ D_H \xi_H$
parasite	$\frac{\partial \bar{z}_P}{\partial t} =$	$-A_P \bar{z}_P$	$+B_P (\bar{z}_H - \bar{z}_P)$	$+ \frac{\sigma_P^2}{2} \nabla^2 \bar{z}_P$	$+ D_P \xi_P$

$\bar{z}(x, t)$: the mean trait at place x and time t

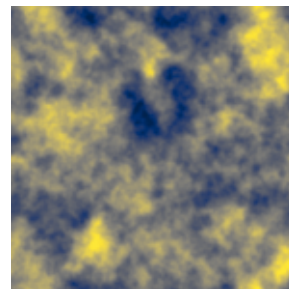


the mean traits
across the landscape:

two random surfaces



host



parasite

**a bivariate
Gaussian random field**

described fully by how alike
two places are, by their distance